Supplementary Information for Spacetime Bounds on Consciousness

Supplementary Note 0 Narrative Guide

This supplementary file is the readable companion to the main paper. It explains the symbols, the models, and the simulation-based checks in friendly, narrative form. Nothing here is new experimental biology. The only empirical anchor is a reanalysis of published primate corpus callosum microstructure from the literature. No new data is collected here.

Supplementary Note 0.1 Claim

The paper separates two frequently conflated ideas. One is that all the ingredients of a conscious moment happen somewhere in a short time window. The other is that there exists an instant inside that window when those ingredients are all true together. The paper calls the first Occurrence and the second CoInstantiation. If you also assume that ingredients have to exchange causal influence within the same window, then there is a maximum spatial size for one unified moment. Formally, the diameter D of the grounded support must satisfy $D \leq \kappa v \theta$. Here θ is the window duration. v is an upper bound on signal speed in the relevant substrate. κ depends only on the communication architecture.

Worked numeric example Suppose $\theta = 20\,\mathrm{ms}$ and $v = 10\,\mathrm{m\,s^{-1}}$. Then $v\theta = 0.2\,\mathrm{m}$. Under hub exchange, meaning $\kappa = 1$, the bound says $D \leq 0.2\,\mathrm{m}$. Under all-to-all exchange, meaning $\kappa = \frac{1}{2}$, the bound says $D \leq 0.1\,\mathrm{m}$. These are order of magnitude numbers, but they show how a time budget becomes a size budget.

Supplementary Note 0.2 Glossary of technical terms

This glossary prioritises terms that are easy to misread across disciplines. If a term looks obvious to you, skip it. If a term looks suspicious, it is probably in here.

Stack Theory objects

- Environment Φ. The space of all possible physical states under consideration.
- Microstate. One element of Φ.
- Program p. A yes or no property represented as the set of microstates where it holds.
- Power set 2^{Φ}. The set of all subsets of Φ. This is the set of all possible programs you could define.

- Vocabulary $\mathfrak{v}$. The finite set of programs a given physical system can actually implement (across different microstates).
- Abstraction layer. An environment plus a vocabulary plus its induced language of compatible bundles.
- Extensional semantics. A way of giving meaning where the meaning of a statement is the set of physical states where it is true. Possible worlds (collections of microstates) given different statements in a system's abstraction layer are asserted, basically.
- Bekenstein bound. A physics upper bound on information content in a finite region. Here it is only used as motivation for treating vocabularies as finite.
- Statement ℓ. A finite bundle of programs that can all be true together.
- Truth set $T(\ell)$. The set of microstates where every program in ℓ is true at once.
- Completion. A more specific statement obtained by adding extra programs.
- Extension set $E(\ell)$. The set of all completions of ℓ inside the vocabulary.
- Weakness $w(\ell)$. How many completions are available. Under the finite vocabulary assumption used here, $w(\ell) = |E(\ell)|$.
- Grounded. Evaluated directly on the base physical state, not only inside an internal abstraction (i.e. the mythical all-knowing eye says it is true, so it must be).

Windowing and unity

- Integration window θ. The time budget available for building one moment.
- Objective time. Time measured by an external physical clock, not the system's own subjective time.
- Occurrence $\mathrm{Occur}(\ell, \sigma)$. Every ingredient program in ℓ becomes true at least once somewhere inside the window.
- CoInstantiation $\mathrm{CoInst}(\ell, \sigma)$. There exists an instant inside the window when the microstate lies in the truth set $T(\ell)$.
- Temporal Gap. Occurrence holds but CoInstantiation fails.

Spacetime and communication architecture

- Grounded support S. The set of physical sites whose grounded ingredients participate in the moment.
- Diameter D. The largest distance between any two sites in S.
- Metric and metric space (X, ρ). A rule for measuring distance that behaves like ordinary distance and obeys the triangle inequality.
- Triangle inequality. The rule that going from a to c directly is never longer than going from a to b to c.

- Undirected graph. A set of nodes connected by edges that do not have a direction.
- Star graph. A hub node connected to every other node.
- Complete graph. Every pair of nodes connected by an edge.
- Speed bound v. An upper bound on how fast causal influence can propagate in the substrate.
- Exchange graph G. The communication pattern between sites.
- Edge-wise round trip. The modelling rule that each edge supports a two-way exchange within one window.
- Hop distance and hop diameter $h(G)$. Graph distance measured in edges and its maximum over pairs.
- Architecture factor κ. Half the hop diameter. In symbols, $\kappa = h(G)/2$.
- Hub exchange. A star graph, so the farthest pair is two hops apart and $\kappa = 1$.
- All-to-all exchange. A complete graph, so every pair is one hop apart and $\kappa = \frac{1}{2}$.

Simulation and statistics terms

- Mechanistic model. A toy message-passing model that makes the inequality concrete.
- Jitter. Random extra delay on top of deterministic travel time.
- Exponential jitter. A delay model where most delays are small but rare delays can be large.
- Gamma distribution. The distribution you get when you add up independent exponential delays. In this paper the sum of two exponential jitters is Gamma with shape 2 and scale $j\theta$.
- Shape parameter. One of the two standard parameters of a Gamma distribution. Shape 2 means you are adding two independent exponential delays.
- Scale parameter. The other standard Gamma parameter. In this paper the scale is $j\theta$, so larger j means larger random delays relative to the window.
- Gamma CDF. The closed form probability that the sum of two exponential jitters stays below a deadline.
- Slack time. The time remaining in the window after subtracting the deterministic travel time. Jitter has to fit inside this slack for integration to succeed.
- CDF. Cumulative distribution function. It converts a random delay into the probability of being under a given deadline.
- Euclidean distance. Straight line distance in ordinary space.
- Chord distance. The straight line distance between two points on a circle.
- Fragmentation transition. A sharp drop in integration success probability as $D/(v\theta)$ increases.
- Sweep. A systematic scan over parameter values. You rerun the same simulation or analysis while changing one or more input parameters.

- Robustness sweep. A sweep where you change nuisance parameters, meaning details that should not matter for the main claim. The goal is to check that the qualitative result does not depend on one special parameter setting.
- x_{50}. The value of x where success probability is 0.5. It is a single-number summary of where the transition happens.
- Monte Carlo replication. One simulated run of the full measurement and test pipeline.
- Margin M_i. The estimated violation margin $\widehat{D}_i - \kappa\widehat{v}_i\widehat{\theta}_i$ for window i. Positive means the bound is violated in that window. Negative means there is spare budget.
- One-sided t test. A standard test that asks whether the mean margin is greater than zero.
- Null hypothesis. The default claim being tested. Here it is that the mean margin is not positive.
- Significance level. The target false rejection probability, often written as α.
- Nominal level. The advertised significance level of a test, which dependence or bad estimators can inflate.
- False rejection rate. The fraction of replications that incorrectly reject when the bound is actually true.
- Type I error. Another name for the false rejection rate.
- Power. The fraction of replications that correctly reject when the bound is truly violated.
- Quantile estimator. A rule that estimates an upper bound by taking a high percentile of samples.
- Percentile. The value below which a given percentage of samples fall. q95 is the 95th percentile.
- q95. The 95th percentile estimator used here for $\widehat{v}$.
- Dependence. Nearby windows can have correlated margins.
- Autocorrelation. A simple way to describe dependence across time.
- AR(1). A minimal dependence model where each sample depends on the previous one plus noise.
- Thinning. Keeping every kth window so that the kept windows are closer to independent.
- Log-log fit. A linear regression after taking logs, which is equivalent to fitting a power law.
- Scaling exponent. The slope in log-log space.
- R^2. The fraction of variance explained by a regression.
- Bootstrap. Resampling the dataset with replacement to estimate stability intervals.
- Ordinary least squares. The standard linear regression fit used for the log-log scaling slopes.

- Pearson correlation. The usual correlation coefficient. In this package R^2 is the squared correlation in log space.
- Phylogeny. The evolutionary relatedness between species, which makes biological data points non-independent.
- Fractional standard deviation. Noise size as a fraction of the measured quantity. This is the unit used for the simulated measurement errors.
- Multiplicative noise. Noise applied by multiplying by a random factor rather than adding a random offset.

Neuroscience and anatomy terms

- Corpus callosum. The large fibre tract that connects the left and right cerebral hemispheres.
- Hemisphere. One half of the cerebrum.
- Interhemispheric. Between the two hemispheres.
- Axon. The long fibre of a neuron that carries electrical signals to other cells.
- Axon diameter. The thickness of an axon. Larger diameters tend to conduct faster.
- Conduction velocity. The speed at which a signal propagates along an axon.
- Conduction time or delay. The time it takes a signal to travel from one site to another.
- Microstructure. Small scale tissue structure. Here it means the axon diameter distribution in the corpus callosum.
- Median. The middle value. Half the observations are above and half are below.
- 95th percentile. The value below which 95 percent of observations fall. Here it is used as a fast fibre proxy.
- Brain mass. Brain weight in grams used as a rough proxy for overall brain size.

Conceptual terms

- Latency budget. The maximum end to end time available for the causal exchanges that are required for one moment to count as unified.
- Bandwidth. Informally, how much information can move through the system per unit time. In this paper it is used as shorthand for effective causal speed relative to system size.
- Causal diamond. The spacetime region that can be mutually reached by signals within the window duration. If the system is bigger than this region allows, you cannot have one co-instantiated moment across the whole support.
- Solid brain and liquid brain. Metaphors from collective intelligence. A solid brain has persistent high-bandwidth wiring. A liquid brain computes through slower interactions among mobile agents such as an ant colony.

Supplementary Note 0.3 What the simulations and analyses test

Mechanistic fragmentation model. This (minimal sanity check) asks whether a concrete message-passing system really does lose its ability to complete all required two-way exchanges at the same $D/(v\theta)$ thresholds predicted by the theorem. The answer is yes in this toy model. Hub exchange collapses near $D/(v\theta) = 1$. All-to-all exchange collapses near $D/(v\theta) = \frac{1}{2}$.

Monte Carlo stress tests. These test the testing protocol. The protocol estimates a conservative speed bound inside each window, computes margins M_i, and tests whether the mean margin is positive. The stress tests check that false rejections stay near the nominal level when the bound is true. They check that power rises quickly when the bound is truly violated. They check that naive non-conservative choices for $\widehat{v}$ can manufacture false violations. They also show how thinning helps when windows are correlated.

Primate literature anchor. Very basic "how might this hold up in reality?". If a conscious moment needs bilateral exchange, then a moment cannot be shorter than the interhemispheric conduction delay. The reanalysis uses published corpus callosum microstructure to estimate those delays across primates and show how they scale with brain size. It does not claim a phylogenetic model or a causal mechanism. It is an order of magnitude anchor for θ.

Supplementary Note 0.4 Figure-by-figure reading guide

What the package actually tested. Each bullet tells you what the plot is, what the axes mean, and what conclusion it supports. Plots are generated by the python file.

- `ttgs_mechanistic_main.pdf.` This is the two panel summary figure for the mechanistic message passing model. Panel (a) plots success probability on the vertical axis against $x = D/(v\theta)$ on the horizontal axis. Success means every required round trip finishes before the window ends. There are two curves, one for hub exchange and one for all-to-all exchange. The vertical reference lines mark the theoretical thresholds $x = 1$ and $x = \frac{1}{2}$. The key thing to notice is that success drops sharply near those thresholds. Panel (b) plots the transition point x_{50} on the vertical axis against the number of sites N on the horizontal axis. x_{50} is the value of x where success probability is 0.5. Different curves correspond to different jitter levels. Solid and dashed line styles separate hub from all-to-all. The key thing to notice is that x_{50} stays close to 1 for hub exchange and close to $\frac{1}{2}$ for all-to-all across the sweep.

- `ttgs_mechanistic_transition.pdf` **and** `ttgs_mechanistic_thresholds.pdf.` These are the single panel versions of the two panels in `ttgs_mechanistic_main.pdf`. They exist so you can inspect each panel without the small font layout constraints of the combined figure. The axes and interpretation are the same as above.

- `ttgs_power_curve.pdf.` This tests statistical power of the falsification protocol under the conservative q95 speed estimator. The horizontal axis is the true ratio $D/(v\theta)$ that generated the synthetic data. The vertical axis is the rejection probability of a one sided t test on the window margins. The dashed vertical line at 1.0 marks the boundary where the bound is exactly tight. Everything to the right of that line is a true violation. In this figure each point uses $n = 40$ windows, 20 within-window speed samples, and multiplicative measurement noise with fractional standard deviations 0.03 on D, 0.03 on θ, and 0.05 on v. Each point is estimated from 2500 Monte Carlo replications at nominal level $\alpha = 0.05$.

- `ttgs_v_estimator_tradeoff.pdf` **and** `ttgs_v_estimator_table.tex`**.** These test how your choice of speed estimator affects false positives and sensitivity. The plot uses the same axes as the power curve. Each curve corresponds to a different estimator applied to the within-window speed samples, such as min, mean, median, q95, or max. Estimators that underestimate v make the bound easier to violate and inflate the false rejection rate. The table reports the same information in numbers for a few representative ratios.

- `ttgs_dependence_thinning.pdf`**.** This tests what happens when windows are correlated. The horizontal axis is the thinning factor k. $k = 1$ means you keep every window. $k = 10$ means you keep every tenth window. The vertical axis is the observed false rejection rate when the bound is actually true. The dashed horizontal line is the nominal $\alpha = 0.05$. The takeaway is that dependence can inflate false positives and thinning pulls them back down.

- `ttgs_power_vs_n.pdf`**.** This tests sample size. The horizontal axis is the number of windows per replication. The vertical axis is rejection probability. One curve shows the Type I error case where the true ratio is 1.00. Another curve shows the power case where the true ratio is 1.01. The takeaway is that more windows buys power quickly for even small violations.

- `ttgs_primate_main.pdf`**.** This is the two panel summary figure for the primate literature anchor. It uses $n = 15$ individuals across 14 primate species from the corrected Phillips et al Table 1. Panel (a) is a log-log scatter plot. The horizontal axis is brain mass in grams. The vertical axis is an implied lower bound $\theta_{\min}$ in milliseconds based on published interhemispheric conduction time proxies. There are two series. One uses a median axon proxy and one uses a fast axon proxy based on the 95th percentile axon diameter. The takeaway is that larger brains have longer minimum bilateral delays. Panel (b) turns the same data into a constraint curve. The horizontal axis is a candidate window duration θ in milliseconds. The vertical axis is the fraction of primate data points whose conduction time exceeds that θ. A small fraction means most species could in principle fit bilateral exchange inside that window. A large fraction means many species could not.

- `ttgs_primate_scaling.pdf` **and** `ttgs_primate_constraint.pdf`**.** These are the single panel versions of the two panels in `ttgs_primate_main.pdf`. They exist for readability. The axes and interpretation are the same as above.

- `ttgs_literature_anchor.pdf`**.** This is a back of the envelope human scale anchor. It combines plausible human brain diameters derived from the same primate table with a published range of callosal conduction velocities. The horizontal axis is implied crossing time in milliseconds for one way travel across the diameter. The plot shows the implied range. It is meant to give you a feel for what D/v looks like in human units.

Supplementary Note 1 Source claims audit

This table lists every numeric input taken from published sources that is used by the manuscript. It also lists the modelling constants used to convert those published values into the plotted anchors. For each row I list the exact variable or constant name used in `ttgs_simulation.py`. All other numerical values in the package are generated by simulation and appear in `ttgs_source_data`.

Input	Where used	Value used	Code name	Source and exact location
Primate corpus callosum table	Supplementary Note 6 and main text primate figures	`ttgs_phillips2015_table1.csv` with brain mass and implied conduction times	`primate_csv` `primate_rows_from_phillips_table1`	Phillips et al 2015, corrected Table 1 as reproduced in the open access correction note (PMCID: PMC4685830). The CSV is a direct transcription of the corrected Table 1. It contains $n = 15$ individuals across 14 species. It uses the corrected species label Macaca maura [1, 2].
Callosal conduction velocity range	Supplementary Note 6 human anchor figure	4.9 to 11.4 m s^{-1}	`CALLOSAL_V_LO_M_PER_S` `CALLOSAL_V_HI_M_PER_S`	Innocenti et al 2022 (PMCID: PMC8752969), Section iii) "Temporal transformations". The sentence that reports mean conduction velocities between 4.9 and 11.4 m s^{-1} across species and areas [3].
Human brain masses	Supplementary Note 6 human anchor figure	1250 g and 1625 g	`human_masses_g`	Phillips et al 2015, corrected Table 1 (PMCID: PMC4685830), rows Homo sapiens F and M [1, 2].
Assumed density for mass to volume conversion	Supplementary Note 6 human anchor figure	1 g cm^{-3}	`DENSITY_G_PER_CM3`	Modelling assumption. I use water density as an order of magnitude conversion from brain mass to volume.
Diameter padding	Supplementary Note 6 human anchor figure	$\pm 5\%$	`DIAMETER_PADDING_FRAC`	Modelling assumption. Padding avoids accidental underestimation of diameter when converting mass to an equivalent sphere diameter.

Supplementary Table 1: Audit table for all non simulated numeric inputs.

Supplementary Note 2 Stack Theory objects and grounding

This section restates the Stack Theory definitions used in the main text. I treat the Stack Theory definitions as fixed. I restate only the parts needed to make the present paper self contained. Stack Theory is developed in my thesis and related work [4, 5, 6].

Definition 1 (Environment)**.** *An environment is a nonempty set Φ of mutually exclusive states.*

Explanation. Φ is the space of all physical states we are willing to talk about. Mutually exclusive means the system is in exactly one of these states at any instant.

Definition 2 (Program and vocabulary)**.** *A program is any set of states $p \subseteq \Phi$. Let $\mathcal{P} := 2^{\Phi}$ denote the set of all programs. A vocabulary $\mathfrak{v} \subseteq \mathcal{P}$ is any set of programs that a system can implement. Unless specified otherwise, vocabularies are finite.*

Explanation. A program is just a property of the world. It is represented as the set of states where that property is true. A vocabulary is the list of properties this particular system can physically represent and use.

Definition 3 (Embodied language and truth set)**.** *The language induced by a vocabulary* $\mathfrak{v}$ *is*

$$L_{\mathfrak{v}} := \{\, \ell \subseteq \mathfrak{v} \mid \bigcap_{p \in \ell} p \neq \emptyset \,\}.$$

Elements $\ell \in L_{\mathfrak{v}}$ *are called statements. Unless specified otherwise, statements are assumed finite. Write*

$$L_{\mathfrak{v}}^{\mathrm{fin}} := \{\, \ell \in L_{\mathfrak{v}} \mid \ell \text{ is finite} \,\}$$

for the finite fragment. If $\mathfrak{v}$ *is finite then* $L_{\mathfrak{v}}^{\mathrm{fin}} = L_{\mathfrak{v}}$*. The truth set of* ℓ *is*

$$T(\ell) := \bigcap_{p \in \ell} p \subseteq \Phi.$$

I use the convention $T(\emptyset) = \Phi$*.*

Explanation. A statement ℓ is a bundle of ingredient programs that can all be true together. The language $L_{\mathfrak{v}}$ is the set of all such compatible bundles that the system could in principle assert. The truth set $T(\ell)$ is the set of physical states where every ingredient in the bundle holds at the same time. The finite fragment just means we only allow bundles with finitely many ingredients. Under the default assumption that the vocabulary itself is finite, every statement is automatically finite.

Definition 4 (Completions, extensions, weakness)**.** *A completion of a statement* $\ell \in L_{\mathfrak{v}}$ *is any* $\ell' \in L_{\mathfrak{v}}$ *with* $\ell \subseteq \ell'$*. The extension set of* ℓ *is*

$$E(\ell) := \{\, \ell' \in L_{\mathfrak{v}} \mid \ell \subseteq \ell' \,\}.$$

The weakness of ℓ *is* $w(\ell) := |E(\ell)|$*. Under the default finite vocabulary assumption this is a natural number.*

Explanation. A completion is what you get when you say more while staying consistent with what you already said. The extension set is the menu of all such further commitments that are still possible within the system's vocabulary. Weakness counts the size of that menu. If weakness is large, the statement still leaves many compatible ways the world could be. If weakness is small, the statement has painted the system into a corner.

Definition 5 (Occurrence and co-instantiation)**.** *Fix a finite window* $\sigma = (\phi_0, \ldots, \phi_\tau)$ *of objective states in* Φ*. A program* p *(a potential fact) occurs in* σ *if* p *is true at least once in the window. Equivalently,* p *occurs when there exists* $k \in \{0, \ldots, \tau\}$ *with* $\phi_k \in p$*. For a statement* $\ell \in L_{\mathfrak{v}}$*, define ingredient-wise satisfaction*

$$\mathrm{Occur}(\ell, \sigma) \iff \forall p \in \ell\, \exists k \leq \tau \quad \textit{with} \quad \phi_k \in p.$$

Define co-instantiated satisfaction

$$\mathrm{CoInst}(\ell, \sigma) \iff \exists k \leq \tau \quad \textit{with} \quad \phi_k \in T(\ell).$$

A temporal gap occurs when $\mathrm{Occur}(\ell, \sigma)$ *holds but* $\mathrm{CoInst}(\ell, \sigma)$ *fails.*

Explanation. Think of σ as the short time slice that is meant to contain one moment. Occurrence means every ingredient appears somewhere in that slice. CoInstantiation means there is at least one instant inside the slice when all ingredients are true together. The temporal gap is the awkward case where each ingredient shows up at some time, but they never overlap. If you prefer continuous time, treat σ as a sampling of the microstate trajectory over $[t, t+\theta]$. The definitions agree because they only ask whether ingredients ever appear and whether any instant contains their overlap.

Remark 1 (Grounding)**.** *Stack Theory separates truth at an abstraction layer from truth in the base environment. In this manuscript I call an ingredient grounded when it is represented as a program in the base environment Φ and evaluated directly on the physical state.*

Explanation. Grounded means we are not letting the system declare victory by using an internal label that is disconnected from the underlying physics. A grounded ingredient is checked against the base state itself.

Supplementary Note 3 Spacetime closure and the architecture factor

I follow the main text and assume a speed bound v on causal propagation through the substrate. I also assume that integration within one moment requires two-way exchange along a specified set of edges.

Definition 6 (Diameter of a finite set of sites)**.** *Let U be a finite subset of a metric space (X, ρ). Define its diameter by*

$$\operatorname{diam}(U) := \max_{u,u' \in U} \rho(u, u').$$

Explanation. $\operatorname{diam}(U)$ is just the largest physical separation between any two sites in U.

Definition 7 (Edge-wise round trip integration graph)**.** *Fix a duration $\theta > 0$ and a finite set of contributor locations U in a metric space (X, ρ). Fix a connected undirected graph $G = (U, E)$. I say U is G integrated within duration θ if for every edge $\{u, u'\} \in E$ there exist travel times $\eta_{u\to u'}, \eta_{u'\to u} \geq 0$ with*

$$\eta_{u\to u'} + \eta_{u'\to u} \leq \theta$$

and the speed bound applies to both directions, meaning

$$\rho(u, u') \leq v\, \eta_{u\to u'} \quad \text{and} \quad \rho(u, u') \leq v\, \eta_{u'\to u}.$$

Explanation. Think of G as a wiring diagram for who must exchange information within one moment. For each required edge, there has to be time for a message to go out and for something to come back within the same integration window of duration θ. The inequalities with v enforce the idea that messages cannot arrive instantaneously because there is a finite speed ceiling in the substrate.

Definition 8 (Hop diameter)**.** *For a connected undirected graph $G = (U, E)$ define the hop distance $d_G(u, u')$ as the length of the shortest path between u and u'. Define the hop diameter*

$$h(G) := \max_{u,u' \in U} d_G(u, u').$$

Explanation. $d_G(u, u')$ is the smallest number of message handoffs you need to get from u to u' using the required edges. $h(G)$ is the worst case of this hop count across the whole graph. It is a pure architecture number. It ignores physical geometry and only remembers how many relays the architecture forces you to use.

Theorem 1 (Graph diameter spacetime bound)**.** *Assume propagation speed at most v. If U is G integrated within duration θ, then for all $u, u' \in U$,*

$$\rho(u, u') \leq \frac{v\theta}{2}\, d_G(u, u').$$

In particular,

$$\operatorname{diam}(U) \leq \frac{v\theta}{2}\, h(G).$$

Proof. Fix $u, u' \in U$ and let $u = u_0, u_1, \ldots, u_k = u'$ be a shortest path in G, so $k = d_G(u, u')$. For each edge $\{u_i, u_{i+1}\}$, edge-wise integration gives travel times with sum at most θ. At least one direction has time at most $\theta/2$. By the speed bound in the faster direction and symmetry of the metric, $\rho(u_i, u_{i+1}) \leq v\theta/2$. The triangle inequality gives

$$\rho(u, u') \leq \sum_{i=0}^{k-1} \rho(u_i, u_{i+1}) \leq k\frac{v\theta}{2} = \frac{v\theta}{2} d_G(u, u').$$

Taking the maximum over u, u' gives the diameter bound. □

Remark 2. *If G is a star graph centred at a hub, then $h(G) = 2$ and the bound becomes* $\operatorname{diam}(U) \leq v\theta$. *If G is complete, then $h(G) = 1$ and the bound becomes* $\operatorname{diam}(U) \leq v\theta/2$. *Any connected graph architecture introduces the factor $h(G)/2$.*

Explanation. Every edge can only cover so much physical distance in one window because signals have limited speed. A path with more hops accumulates more of these hop limits. So the largest possible diameter is controlled by how many hops the farthest pair needs, which is exactly the hop diameter.

Supplementary Note 4 Mechanistic model details

The mechanistic model in the main text evaluates the exact probability of successful integration under independent exponential jitter. I include the single panel plots used to build the main text figure.

Explanation. This model is a deliberately simple stress test. It turns the bound into a probability question. Given a deadline θ, how often do all the required messages manage to go out and come back in time. As the ratio $D/(v\theta)$ grows, the answer falls off a cliff. That cliff is the fragmentation transition reported in the main text.

Supplementary Note 4.1 Model definition and closed form success probabilities

Let D be the physical diameter of the circle and let θ be the window duration. Let v be the signal speed ceiling. Define the dimensionless diameter ratio

$$x := \frac{D}{v\theta}.$$

Each directed message over distance d has travel time

$$\tau_{\rightarrow} = \frac{d}{v} + \xi, \qquad \xi \sim \mathrm{Exp}(\text{mean } j\theta).$$

A two-way exchange over the same distance has round trip time

$$\tau_{\mathrm{rt}} = \frac{2d}{v} + \xi_1 + \xi_2, \qquad \xi_1, \xi_2 \text{ independent.}$$

Write $S := \xi_1 + \xi_2$. Because S is the sum of two independent exponential random variables, it is Gamma distributed with shape 2 and scale $j\theta$. For any slack time $s \geq 0$,

$$\Pr[S \leq s] = 1 - \exp\Big(-\frac{s}{j\theta}\Big)\Big(1 + \frac{s}{j\theta}\Big).$$

If $s < 0$ then the probability is 0 because the deterministic travel time already exceeds the window.

Hub exchange Each contributor sits at radius $D/2$ so its distance to the hub is $d = D/2$. The deterministic round trip is $2d/v = D/v = x\theta$. So a hub round trip succeeds exactly when $S \leq \theta(1-x)$. Let $F(\cdot)$ denote the Gamma CDF above after scaling by θ. Then for $x < 1$,

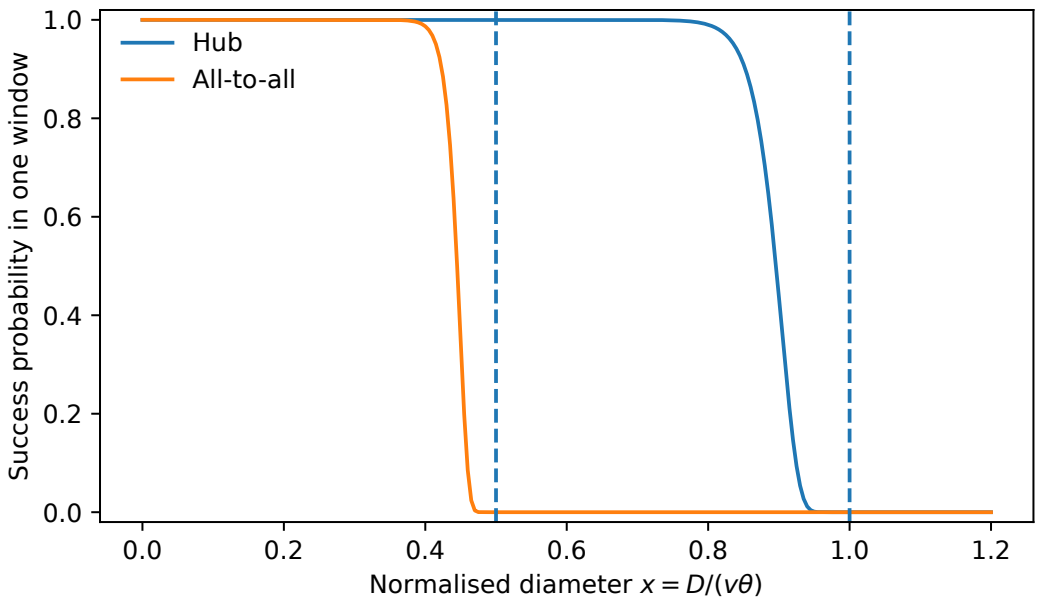
$$p_{\mathrm{hub}}(x) = F(1-x)^N.$$

For $x \geq 1$ we have $p_{\mathrm{hub}}(x) = 0$.

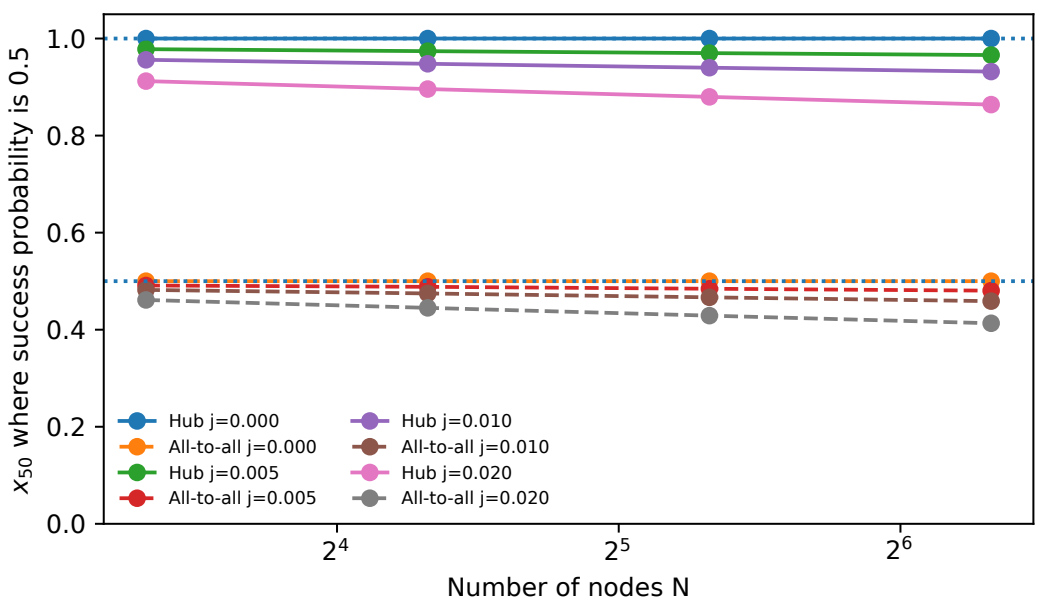
All-to-all exchange Two sites separated by angular gap Δ have chord distance $d = D\sin(\Delta/2)$. So their deterministic round trip is $(2D/v)\sin(\Delta/2) = 2x\sin(\Delta/2)\,\theta$. The round trip succeeds when $S \leq \theta(1 - 2x\sin(\Delta/2))$. If any pair has negative slack, success is impossible for the whole window. Otherwise, under the independent-message assumption, the all-to-all success probability is the product over pairs

$$p_{\mathrm{all}}(x) = \prod_{i<j} F\big(1 - 2x\sin(\Delta_{ij}/2)\big).$$

Explanation. The only moving parts are geometry and a deadline. The geometry determines how much deterministic travel time you spend just moving signals around. The deadline is the window θ. Jitter is the random tax you pay on top. The Gamma formula above is the probability that two independent exponential delays fit inside the slack left after you pay the deterministic travel time.



Supplementary Fig. 1: Success probability versus $x = D/(v\theta)$ for hub and all-to-all exchange at $N = 20$ and jitter mean $j = 0.02$. Success means that every required two-way exchange finishes before the window ends. As x increases, the time budget gets tighter and the success probability drops rapidly.



Supplementary Fig. 2: Robustness sweep of the transition location. For each node count N and jitter level j, I compute the success curve and record x_{50}, the value of x where success probability is 0.5. Solid curves are hub exchange and dashed curves are all-to-all exchange. Horizontal reference lines mark the theoretical thresholds $x = 1$ and $x = \frac{1}{2}$. The main point is that x_{50} stays close to these values across the sweep.

Narrative reading of the sweep. Each point on a curve is one full rerun of the model at a chosen N and j. The vertical axis x_{50} is the tipping point where success falls to 50%. Curves near 1 mean the hub architecture fails only when the window is tight enough that crossing the diameter consumes almost the whole window. Curves near $\frac{1}{2}$ mean the all-to-all architecture fails earlier, when crossing the diameter consumes only half the window, because it requires stricter within-window round trips. The weak dependence on N is the robustness result. It says the tipping point is not sensitive to the number of sites in the tested range.

Supplementary Note 5 Statistical protocol and stress tests

This section documents the falsification pipeline used in the main text stress tests. All results are generated by `ttgs_simulation.py`. Source data for every figure is written into the `ttgs_source_data` directory.

Supplementary Note 5.1 Margin definition and test

I test the inequality $D \leq \kappa v \theta$ using the estimated margin

$$M_i = \widehat{D}_i - \kappa \widehat{v}_i \widehat{\theta}_i$$

across windows $i = 1, \ldots, n$. The null is that the mean margin is non positive. I reject using a one-sided t test on $\{M_i\}$. In the Monte Carlo experiments I set $\kappa = 1$, so the simulated ratio axis is $D/(v\theta)$. For other communication architectures, reinterpret the same curves by plotting $D/(\kappa v \theta)$.

Explanation. Each window gives a single number M_i. If M_i is negative, the bound holds with slack in that window. If M_i is positive, that window violates the bound. The one-sided t test asks whether the average margin across windows is greater than zero. This is conservative in the intended direction. A rejection is hard to obtain if $\widehat{v}$ and $\widehat{\theta}$ are upper bounds and if $\widehat{D}$ is not overestimated.

Explanation. translation of the reporting terms used below A *false rejection rate* is the fraction of Monte Carlo replications that reject the null when the true ratio is $D/(v\theta) = 1.00$. This is the Type I error rate of the pipeline. A *power* value is the fraction of replications that reject when the true ratio is larger than 1.00, for example 1.03. Each plotted rejection probability is computed as the number of rejections divided by the number of independent Monte Carlo replications.

Supplementary Note 5.2 Why v must be a relatively conservative upper bound

In the inequality, v is a speed *limit* rather than an average. If $\widehat{v}$ underestimates the relevant speed bound, then M is biased upward and false violations appear. So I treat v as sampled within each window and I use a high quantile estimator. The default rule used here is `q95`. It sets $\widehat{v}_i$ to the empirical 95th percentile of the n_{samples} speed samples drawn within-window i. This is conservative in the sense that it overestimates typical speeds while still respecting that v should behave like a ceiling rather than an average.

Explanation. q95 is a compromise between using the maximum, which is noisy, and using the mean, which is too slow for a speed limit. It is asking for a high but not extreme value, so that you do not accidentally underestimate the fastest relevant propagation.

The table below compares estimators. It reports the false rejection rate when the true ratio is 1.00 and the power when the true ratio is 1.03.

v estimator	Type I at ratio 1.00	Power at ratio 1.03
Minimum of samples	1.000	1.000
Mean of samples	0.047	0.995
Median of samples	0.046	0.993
95th percentile of samples	0.000	0.000
Maximum of samples	0.000	0.000

Supplementary Table 2: Estimator tradeoff table from Monte Carlo stress tests.

Supplementary Note 5.3 Power and error control figures

Simulation settings Unless otherwise stated, the stress tests use $n = 40$ windows per replication, 20 within-window speed samples to estimate $\widehat{v}_i$, and multiplicative Gaussian noise with fractional standard deviations $\sigma_D = 0.03$, $\sigma_\theta = 0.03$, and $\sigma_v = 0.05$.

Quantity	Value
Windows per replication n	40
Within-window v samples per window	20
Noise on $\widehat{D}$	$\sigma_D = 0.03$ fractional stdev
Noise on $\widehat{\theta}$	$\sigma_\theta = 0.03$ fractional stdev
Noise on $\widehat{v}$	$\sigma_v = 0.05$ fractional stdev
Nominal test level	$\alpha = 0.05$
Dependence model in thinning test	AR(1) with $\phi = 0.8$
Thinning factors tested	$k \in \{1, 2, 5, 10\}$

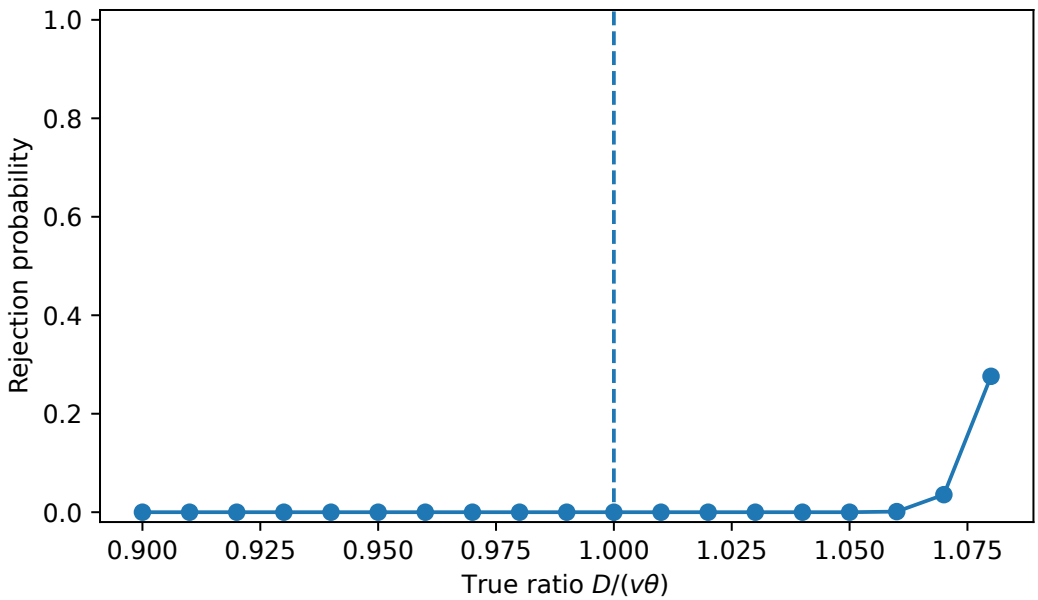
Supplementary Table 3: Default Monte Carlo settings used in the stress tests unless otherwise stated.

These settings describe how noisy the simulated measurements are. They are fractional errors. For example, $\sigma_D = 0.03$ means the diameter estimate is perturbed by noise on the order of three percent.

The power curve figure `ttgs_power_curve.pdf` uses 2500 Monte Carlo replications per point. The estimator comparison figure `ttgs_v_estimator_tradeoff.pdf` and the thinning figure `ttgs_dependence_thinning.pdf` use 2000 replications per point. The power-versus-n figure `ttgs_power_vs_n.pdf` uses 1500 replications per point.

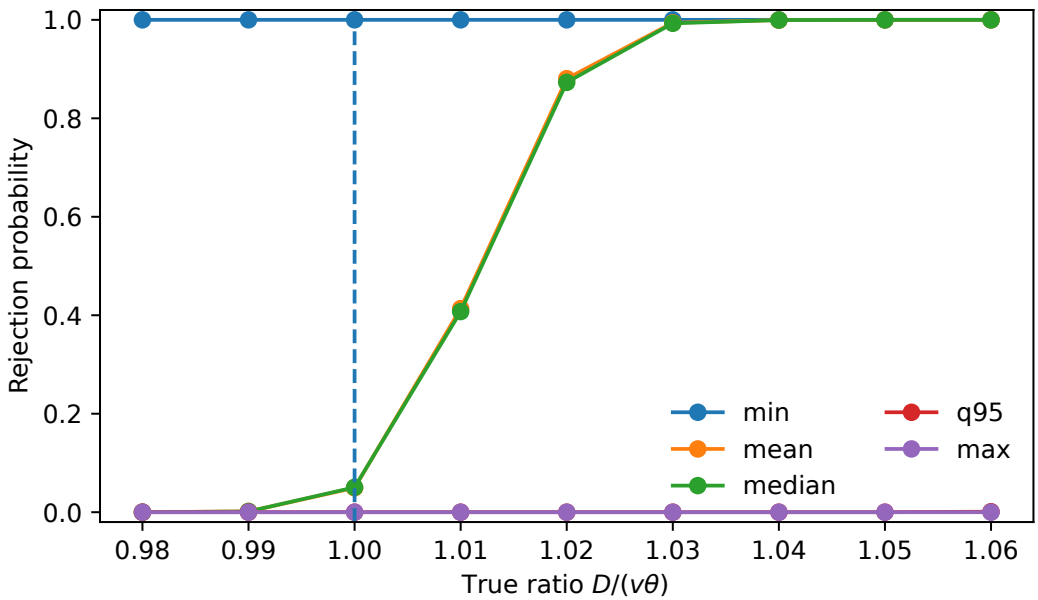
Dependence control by thinning The dependence thinning figure `ttgs_dependence_thinning.pdf` simulates an AR(1) process with autocorrelation parameter $\phi = 0.8$. I then thin the sequence by keeping every kth window, for $k \in \{1, 2, 5, 10\}$, and report the resulting false rejection rate. The point is not that AR(1) is the true dependence model. The point is that mild dependence can inflate Type I error, and thinning is an easy way to reduce that inflation.

AR(1) is a minimal model of correlation across time. It says that if one window happens to have a positive noise shock, the next window tends to inherit some of it. Thinning is the blunt tool that fixes this. Instead of treating every adjacent window as a new independent test case, you keep every kth window and discard the rest. This reduces how much the kept windows overlap and makes standard tests less overconfident.



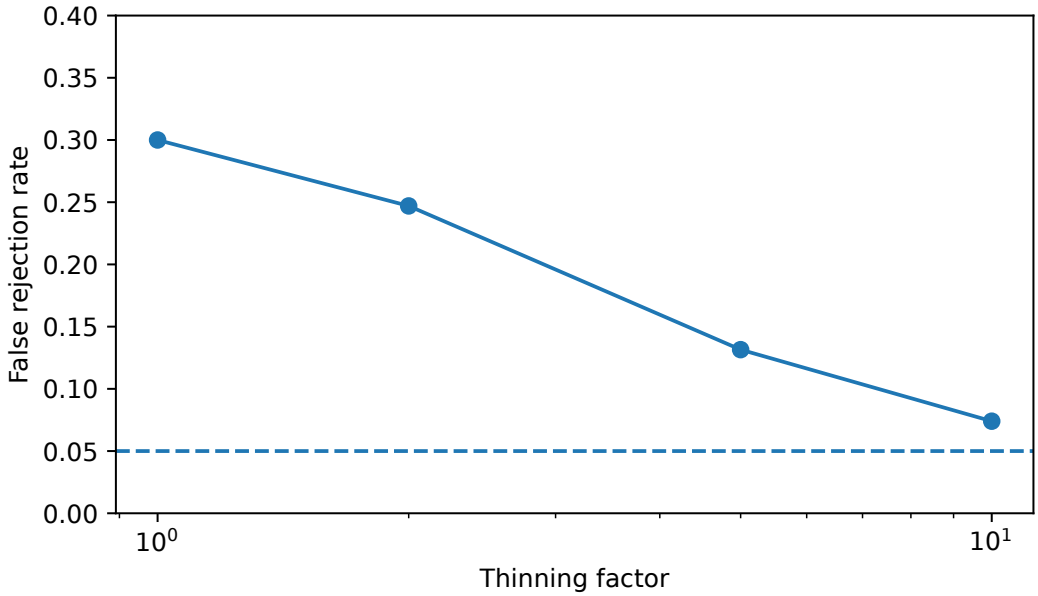
Supplementary Fig. 3: Rejection probability versus true ratio $D/(v\theta)$ when v is estimated conservatively by the 95th percentile.

Narrative reading. This plot checks whether the falsification pipeline behaves sensibly on synthetic data where the truth is known. The horizontal axis is the true ratio $D/(v\theta)$ used to generate the Monte Carlo data. Values below 1 mean the bound is actually true. Values above 1 mean the bound is actually violated. The vertical axis is the fraction of Monte Carlo replications that end in a rejection, meaning the pipeline declares a violation. The dashed vertical line marks the knife-edge case $D/(v\theta) = 1$. A trustworthy pipeline stays near the nominal level at that line and climbs upward as you move to the right.



Supplementary Fig. 4: Rejection probability curves for different v estimators.

Narrative reading. Each curve repeats the same stress test while changing only how the speed limit v is estimated within each window. If an estimator tends to underestimate v, the bound looks too hard to satisfy and the pipeline rejects too often, which creates false positives. If an estimator tends to overestimate v, the bound looks too easy to satisfy and the pipeline misses real violations. The desired behaviour is a curve that stays low at the dashed line and rises quickly just to the right of it. This is why the main text uses a high quantile rule such as q95.



Supplementary Fig. 5: False rejection under AR(1) dependence and thinning.

Narrative reading. This plot shows what happens if windows are correlated across time but you pretend they are independent. The horizontal axis is the thinning factor k. $k = 1$ means you keep every window. $k = 10$ means you keep every tenth window. The vertical axis is the observed false rejection rate when the bound is actually true. The dashed horizontal line is the nominal level $\alpha = 0.05$. The main point is that dependence can inflate false positives and thinning pulls them back down.

Supplementary Note 6 Primate dataset plots

I include the single panel versions of the primate plots. The dataset contains $n = 15$ individuals. The log-log fits are descriptive. The individuals are not independent because of shared phylogeny and shared measurement pipelines. I therefore use the fits as a scaling sanity check rather than as a phylogenetic comparative claim.

This section is not trying to do careful evolutionary statistics. It is checking whether the numbers are even in the right ballpark. If bigger brains have longer callosal delays, then any theory that needs fast global exchange should feel the squeeze as brains scale up.

Supplementary Note 6.1 How the scaling exponents and intervals are computed

Let m_i be brain mass in grams for individual i and let t_i be the reported callosal conduction time proxy in milliseconds. I fit a log-log linear model

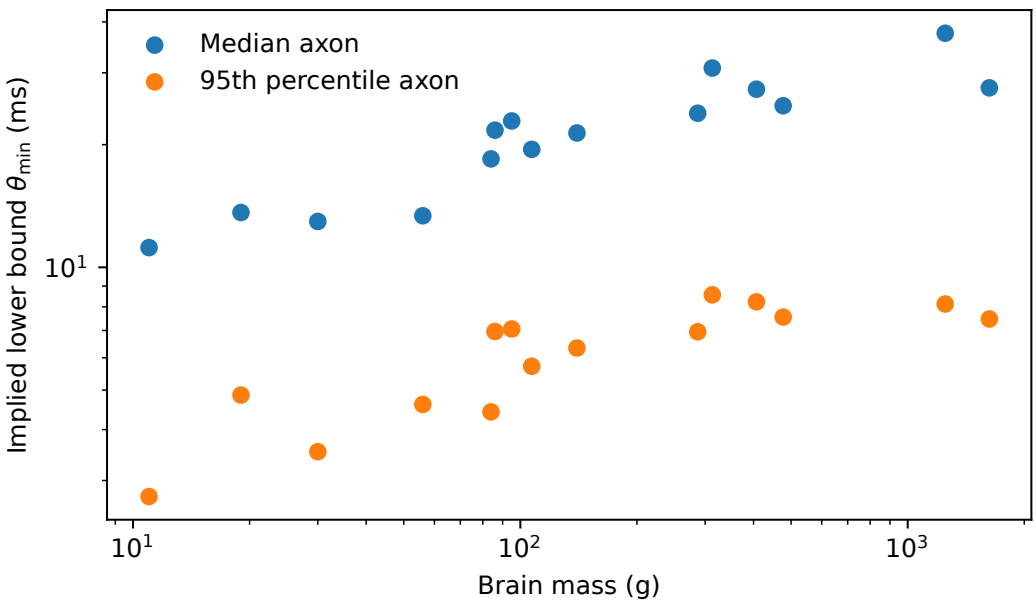
$$\log t_i = a + b \log m_i + \varepsilon_i.$$

A log-log fit is what you do when you suspect a power law. If t scales like m^b, then taking logs turns it into a straight line with slope b. So b is the exponent that says how fast conduction delay grows with brain mass.

The slope b is reported in the main text as the scaling exponent. The reported R^2 is the squared Pearson correlation returned by ordinary least squares on the log transformed data.

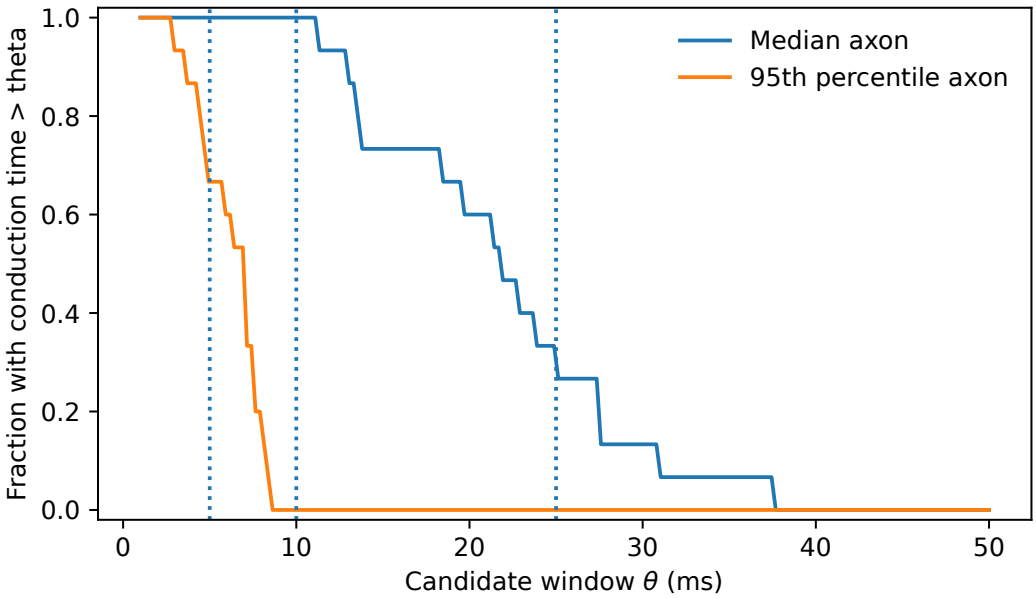
To summarise uncertainty without making a phylogenetic claim, I use a nonparametric bootstrap over the n individuals. I resample the n pairs (m_i, t_i) with replacement 5000 times, refit the slope each time, and report the 2.5th and 97.5th percentiles of the bootstrap slope distribution as the interval. This is an internal stability check, not a phylogenetic comparative inference.

The bootstrap is a brute force way to see whether the exponent is sensitive to any single data point. It repeatedly resamples the individuals and refits the slope. If the slope stays similar across resamples, the scaling pattern is at least internally stable. It does not magically make the individuals independent.



Supplementary Fig. 6: Scaling of implied lower bounds from corpus callosum conduction times.

Narrative reading. Each dot is one primate individual from the corrected Phillips et al dataset. The horizontal axis is brain mass. The vertical axis is an implied minimum window $\theta_{\min}$ in milliseconds based on published interhemispheric conduction delay proxies. A point at $\theta_{\min} = 20\,\text{ms}$ means a window shorter than 20 ms would not be compatible with that individual's reported bilateral delay proxy, even before accounting for extra within-window coordination. The two series use two different proxies, one based on a median axon and one based on a fast axon. The main point is the upward trend. Larger brains sit higher, so their bilateral latency floor is longer.

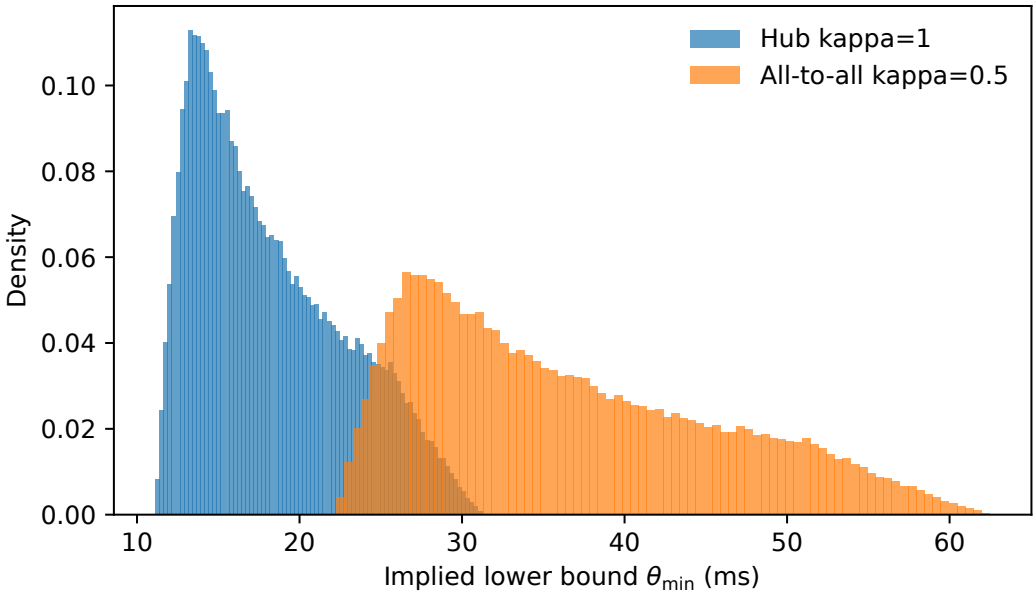


Supplementary Fig. 7: Fraction of individuals exceeding a candidate window θ.

Narrative reading. This plot turns the same primate data into a constraint curve for a candidate window. Pick a candidate window duration θ on the horizontal axis. The vertical axis gives the fraction of individuals whose bilateral delay proxy exceeds that θ. Those individuals could not complete even one cross-hemisphere delay inside that window in the best case. The vertical dotted lines mark three reference windows, 5 ms, 10 ms, and 25 ms, used only as visual anchors.

The primate plots use Table 1 from Phillips et al [1] including the published correction [2]. For the anchor I take a human brain diameter range derived from the human brain masses in Table 1. I convert mass to an equivalent sphere diameter by assuming density $1\,\mathrm{g\,cm^{-3}}$ and then pad the result by $\pm 5\%$ to avoid underestimating diameter. I combine this with the range 4.9 to $11.4\,\mathrm{m\,s^{-1}}$ for mean callosal conduction velocities reported across species and areas by Innocenti et al [3].

This figure is a back-of-the-envelope scale anchor. It translates two published facts into a rough crossing time. A bigger diameter and a slower conduction speed both push the minimum feasible integration window upward.



Supplementary Fig. 8: A human scale anchor built from brain diameter estimates and published callosal conduction velocities.

Narrative reading. This histogram is a rough human scale anchor built from published ranges rather than from new measurements. Each random draw picks a plausible brain diameter and a plausible callosal conduction speed from the cited ranges. It converts them into an implied minimum feasible window $\theta_{\mathrm{min}} = D/(\kappa v)$. The two colours correspond to two architecture factors κ that represent different within-window exchange graphs. The spread of each histogram is uncertainty from the input ranges. The bulk location is the order of magnitude window suggested by those inputs.

Supplementary References

[1] Kimberley A. Phillips, Cheryl D. Stimpson, Jeroen B. Smaers, Mary Ann Raghanti, Bob Jacobs, Aleksandar Popratiloff, Patrick R. Hof, and Chet C. Sherwood. The corpus callosum in primates: processing speed of axons and the evolution of hemispheric asymmetry. *Proceedings of the Royal Society B: Biological Sciences*, 282(1818):20151535, 2015. doi:10.1098/rspb.2015.1535.

[2] Kimberley A. Phillips, Cheryl D. Stimpson, Jeroen B. Smaers, Mary Ann Raghanti, Bob Jacobs, Aleksandar Popratiloff, Patrick R. Hof, and Chet C. Sherwood. Correction to The corpus callosum in primates: processing speed of axons and the evolution of hemispheric asymmetry. *Proceedings of the Royal Society B: Biological Sciences*, 282(1819):20152620, 2015. doi:10.1098/rspb.2015.2620.

[3] Giorgio M. Innocenti, Ingo Vahlsing, and Roberto Caminiti. The functional characterization of callosal connections. *Progress in Neurobiology*, 208:102186, 2022. doi:10.1016/j.pneurobio.2021.102186.

[4] Michael Timothy Bennett. *How To Build Conscious Machines*. PhD thesis, The Australian National University, 2025. URL: https://hdl.handle.net/1885/733782452, doi:10.25911/ta58-p428.

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